

Research Article

Educational Video of Mammography Procedure

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Abstract: A lack of knowledge regarding the procedure causes many women to feel nervous and uneasy about getting a mammogram. This can lead to avoidance of screenings, potentially delaying the detection of breast cancer. Traditional written materials may not effectively address the concerns that women have about the procedure, contributing to their unease. Furthermore, it is not effective in reaching and educating illiterate women or those who have limited access to health information. The goal of this innovation project is to develop an educational video that educates and prepares women for the procedure providing clear and comprehensive information about mammograms, including what to expect and how to prepare for it. The video is created using modern technology involving a team of experts in the breast cancer field, video production, and instructional design. This will enhance engagement and education through the use of interactive elements in video to empower women with the knowledge and confidence to undergo mammograms, ultimately improving breast cancer detection rates. The video innovation has the potential to be commercialized and used as a resource for healthcare providers, mammography facilities, and women's health organizations for education purposes.

Keywords: Breast cancer; Mammography; Educational video.



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1. INTRODUCTION

Most women feel nervous and uneasy about undergoing mammography due to a lack of knowledge regarding the procedure itself. Mammography is a gold standard for detecting breast cancer at its earliest stages. However, women may not know exactly what happens during a mammogram, leading to anxiety and uncertainty. They might believe that mammograms cause pain or harm. Some women worry about exposure and embarrassment during the examination (Meissner et. al. 2007). These factors contributed to a sense of apprehension and hesitancy among women considering mammography.

However, educational interventions, such as simulation models and health education with media leaflets, are effective in improving knowledge and attitudes toward mammogram screening among women (Karimian et. al., 2022; Idrees et. al., 2023). These findings emphasize the importance of exploring alternative educational strategies in empowering women to take control of their breast health and overcome barriers to screening. Thus, this innovation project focuses on creating an interactive instructional video designed to educate and prepare women for the mammogram process.

2. METHOD & MATERIAL

The video aims to present detailed explanations of each stage of the mammogram process. The style, scope, storyboarding, and scriptwriting were considered during the pre-production stage, ensuring that viewers fully comprehend the procedure before attending their appointment, during, and after the mammogram procedure. The video will tackle common worries and misconceptions about mammograms, assisting women to better understand the risks and benefits involved.

The preparation for filming during production was set in one private hospital focusing on the light, sound, and demonstration equipment for optimizing video production in a non-studio setting. The mammogram demonstration especially the compression of breast tissue was done using breast phantom made of 3B SKINlike™ (3B Scientific, Germany) high-quality silicone. The film was then modified in post-production, adding an audio commentary and interactive features to highlight important areas, such as shapes, arrows, and drawing motion.

The video was created shorter and used a conversational tone to engage viewers. A survey using Google Forms was distributed to respondents during a breast cancer awareness campaign. The survey's objective is to gather comments and feedback.



Figure 1. The screenshot of the educational video demonstrates the mammogram technique in detail on breast phantom made of 3B SKINlike™ (3B Scientific, Germany) high-quality silicone.

3. FINDINGS

A survey on the effectiveness of the educational video was carried out among 54 people during one campaign to raise awareness about breast cancer. The age group with the largest percentage (33.3%) is 18 to 30 years old. This is followed by 30 to 40 years old (24.1%), 40 to 50 years old (22.2%), age over 50 (18.5%), and age under 18 (1.9%). The response in detail and the effect of the educational video on their view and desire to have a mammogram (MMG) are shown in Figure 2.

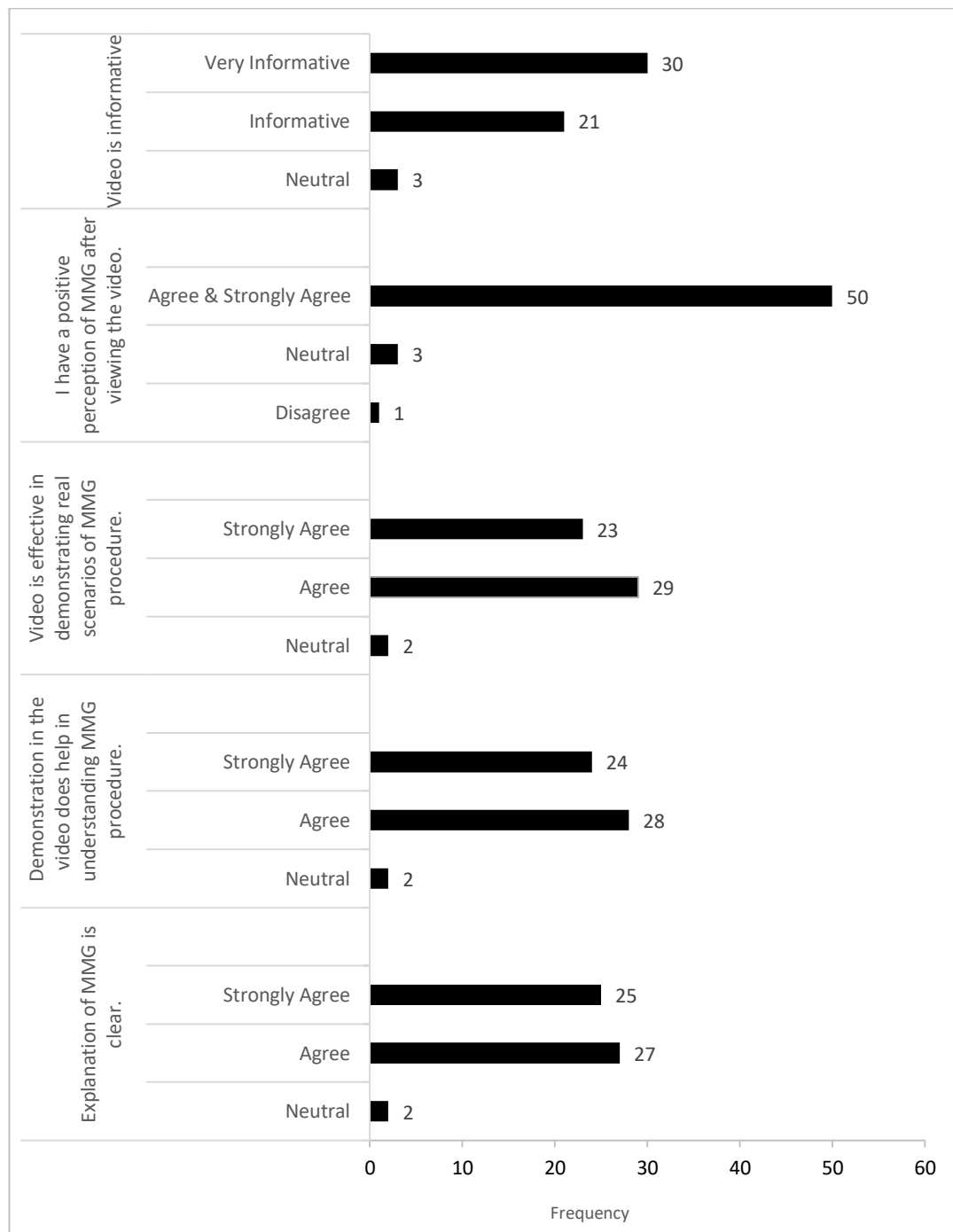


Figure 2. Feedback and the impact of the educational video on perception and motivation to undergo MMG.

4. DISCUSSION

4.1 Uniqueness

This video is different than any other educational video since it provides thorough descriptions of every step involved in getting a mammogram. It presents the real scenario of how the women will experience during their mammogram appointment. It also shows the overview of the procedure room, the mammogram machine, and how the procedure will be done. The compression techniques that most women are afraid of were demonstrated using breast phantoms that have the same tissue composition

as human breast tissue. This will increase confidence, empower women to make informed decisions about their breast health, and reduce the number of individuals avoiding essential preventive care measures.

4.2 Benefits

This type of educational video can be particularly useful for women who are less literate or have limited access to health information, as it provides a visual and interactive way to learn about mammograms. Videos can effectively convey complex information in a visually engaging manner, allowing learners to grasp key points quickly and retain information easily. Videos can reach large audiences regardless of geographical location and they offer the opportunity to pause, rewind, and repeat sections as needed, fostering a deeper understanding and retention of information besides enabling people to learn at their own pace and convenience (Karimian et. al., 2022).

4.3 Commercialization

The video can be sold or licensed to healthcare providers, educational institutions, and other organizations interested in promoting early detection of breast cancer. The video can also be monetized through advertising or sponsorship opportunities. Given the importance of the content in promoting women's health, it can attract sponsors or advertisers interested in supporting such initiatives. It can also be used as a valuable resource for professional training and resource for breast cancer awareness campaigns and events.

5. CONCLUSION

Educational video on mammography has a high potential to empower women with the knowledge and confidence to undergo mammograms. It also has the potential to be commercialized through various channels making it a valuable resource for both healthcare professionals and the general public.

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